

Weak Banach-Space Groups Are Fattening Exactly in Finite Dimension

A full candidate answer to the weak-topology branch of Question 1 in arXiv:2104.02695

Automated mathematics run `fa_banach_001`, lane 5

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Status. Candidate full solution to the explicit weak-topology branch of Open Question 1, likely valid and awaiting human review. The broader parts of Question 1 concerning arbitrary Hausdorff or Polish groups are not claimed.

1 Source question and definitions

The source is D. Ravasini, *Compactivorous Sets in Banach Spaces*, arXiv:2104.02695, later published in *Proceedings of the American Mathematical Society* **150** (2022), 2121–2129 [1]. On page 8, Open Question 1 asks in particular what can be said about weak topologies of Banach spaces without the Schur property. The entire open-question section is reproduced below; the present packet answers precisely the weak-topology clause of item (1).

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6. OPEN QUESTIONS

The following questions arose naturally during the writing of this paper.

- (1) What other Hausdorff topological groups are fattening? What can be said about the weak topologies of Banach spaces without the Schur property or about Polish groups in general?
- (2) Esterle, Matheron and Moreau were the first to ask in [7] whether every closed, convex set which is not Haar null in a separable Banach space must be compactivorous. We observe that it is enough to ask the same question in ℓ_1 . Indeed, let Y be a closed subspace of ℓ_1 , let $C \subset \ell_1/Y$ be a closed, convex subset and let $\pi : \ell_1 \rightarrow \ell_1/Y$ be the quotient map. Provided that the answer is affirmative, one obtains the following chain of implications: C is not Haar null $\Rightarrow \pi^{-1}(C)$ is not Haar null $\Rightarrow \pi^{-1}(C)$ is compactivorous $\Rightarrow C$ is compactivorous. Using invariance under isomorphisms, the result would then become true in all separable Banach spaces.

Figure 1: Open Questions 1 and 2 in the source paper, page 8.

For a Hausdorff topological group G , a set $E \subseteq G$ is *compactivorous* if, for every compact $K \subseteq G$, there are an open set $V \subseteq G$ and $g, h \in G$ such that

$$K \cap V \neq \emptyset, \quad g(K \cap V)h \subseteq E.$$

It is *strongly compactivorous* if there is a closed neighbourhood U of the identity such that, for every compact $K \subseteq U$, some $g, h \in G$ satisfy $gKh \subseteq E$. The group is *fattening* if every compactivorous

subset is strongly compactivorous. For the additive group of a Banach space, the two-sided translate gAh is simply one translate of A .

The source proves that an infinite-dimensional Banach space with the Schur property is nonfattening in its weak topology. It uses the Schur property to turn weakly compact sets into norm-compact sets. The theorem below removes that hypothesis completely.

2 Proof intuition

There are two independent scales in a weakly topologised Banach space. Weakly compact sets have small pieces in the norm metric: the classical norm-fragmentation theorem says that every weakly compact set has a nonempty relatively weak-open subset of arbitrarily small norm diameter. Translating one point of such a piece to the origin puts the whole piece in the norm unit ball. Thus the norm unit ball is compactivorous even though it has empty weak interior in the typical infinite-dimensional case.

On the other hand, a basic weak neighbourhood restricts only finitely many functionals. Their common kernel is an infinite-dimensional direction on which the neighbourhood is unbounded. It therefore contains a two-point compact set of diameter larger than the unit ball's diameter. No translation can fit that set into the unit ball, so the unit ball is not strongly compactivorous. Finite-dimensional spaces fall on the opposite side because weak and norm topologies coincide and the resulting group is locally compact.

3 The fragmentation input

A metric d *fragments* a topological space T if, for every nonempty $A \subseteq T$ and every $\varepsilon > 0$, there is an open $O \subseteq T$ such that $A \cap O \neq \emptyset$ and $\text{diam}_d(A \cap O) < \varepsilon$.

Lemma 1 (Weak-compact norm fragmentation). *Let X be a Banach space and let $K \subseteq X$ be weakly compact. For every $\varepsilon > 0$ there is a weakly open set $V \subseteq X$ such that*

$$K \cap V \neq \emptyset, \quad \text{diam}_{\|\cdot\|}(K \cap V) < \varepsilon.$$

This is the standard theorem that weakly compact subsets of Banach spaces are fragmented by the norm; see Jayne–Namioka–Rogers [2]. Their theorem has no separability or Schur hypothesis. Applying the definition of fragmentation to $A = K$ gives the displayed formulation. This is the only non-elementary external result used below.

4 Main theorem

Theorem 2. *Let X be a real or complex Banach space, and give its additive group the weak topology $\sigma(X, X^*)$. Then the following are equivalent:*

1. $(X, \sigma(X, X^*))$ is fattening;
2. X is finite-dimensional.

More precisely, whenever X is infinite-dimensional, its closed norm unit ball B_X is compactivorous in $(X, \sigma(X, X^))$ but is not strongly compactivorous.*

Proof. Suppose first that X is infinite-dimensional. Let $K \subseteq X$ be any nonempty weakly compact set. Apply Lemma 1 with $\varepsilon = 1$ to obtain a weakly open $V \subseteq X$ such that $K \cap V$ is nonempty and has norm diameter less than 1. Choose $k_0 \in K \cap V$. Then

$$-k_0 + (K \cap V) \subseteq B_X,$$

because $\|k - k_0\| < 1$ for every $k \in K \cap V$. Since K was arbitrary, B_X is compactivorous in the weak topology.

We next show that B_X is not strongly compactivorous. Let U be an arbitrary weak neighbourhood of 0; the argument therefore also applies when U is closed. Some basic weak neighbourhood

$$W = \{x \in X : |x_j^*(x)| < \delta \text{ for } j = 1, \dots, n\}$$

is contained in U , for suitable $x_1^*, \dots, x_n^* \in X^*$ and $\delta > 0$. The subspace

$$Y = \bigcap_{j=1}^n \ker x_j^*$$

has finite codimension and is nonzero (indeed infinite-dimensional), so $Y \subseteq W$ contains a vector z with $\|z\| > 1$. The two-point set $K = \{-z, z\}$ is weakly compact and lies in U . Every translate of K has diameter $2\|z\| > 2$, whereas every subset of B_X has diameter at most 2. Consequently no translate of K lies in B_X . Since this happens inside every weak neighbourhood U of 0, the set B_X is not strongly compactivorous. Hence $(X, \sigma(X, X^*))$ is not fattening.

Finally, suppose X is finite-dimensional. The weak and norm topologies coincide. For completeness, let $E \subseteq X$ be compactivorous and apply the definition to a compact closed ball C with nonempty interior. There are an open V and a translate $t + (C \cap V) \subseteq E$, with $C \cap V \neq \emptyset$. Since $\text{int } C$ is dense in C , the set $V \cap \text{int } C$ is nonempty. Thus $C \cap V$ contains a translate $c + W$ of an open neighbourhood W of 0, and E contains $t + c + W$. Choose a closed neighbourhood $U \subseteq W$. For every compact $K \subseteq U$, the fixed translate $t + c + K$ lies in E . Hence E is strongly compactivorous, so X is fattening. (This is the elementary finite-dimensional instance of the source paper's observation that locally compact groups are fattening.) $\square$

5 Verification notes

The proof uses exactly one imported geometric fact, Lemma 1. Its quantifiers match the compactivorous definition: the relatively weak-open piece is the intersection of K with an ambient weakly open set, and the theorem applies to arbitrary weakly compact K , without metrizability or separability. The remainder is elementary. Translation preserves diameter; a basic weak neighbourhood contains the common kernel of finitely many functionals; and that kernel is nonzero in infinite dimension. The witness $\{-z, z\}$ is finite, hence compact in every Hausdorff topology on X .

No numerical or symbolic experiment is part of the proof. A separate verification report in the packet records a dependency and definition audit. The main human-review target is the precise norm-fragmentation theorem and its application to the source paper's group-theoretic definition.

6 Scope and novelty check

This is a full answer to the explicit weak-topology branch of Open Question 1: there is no additional infinite-dimensional class to find, because every infinite-dimensional weak Banach-space group is

nonfattening. It does not answer the other clauses about arbitrary Hausdorff groups or Polish groups, and it does not address Open Question 2 about Haar-null closed convex sets.

The bounded novelty search was performed on 9 August 2026. The run indexes were searched for arXiv:2104.02695, **compactivorous**, **fattening**, and the weak-topology/Schur terminology. Web and arXiv searches used the exact source title and the phrases “weak topologies of Banach spaces”, “fattening”, “without the Schur property”, and close combinations with “compactivorous”. The searches surfaced the source article, its journal version, and later work on Haar-null convex sets, but no later resolution of this weak-topology question. This is evidence only, not a claim of exhaustive bibliographic novelty; specialist review remains necessary.

7 Human-review recommendation

Review as a focused full solution to the weak-topology subquestion. First confirm that the cited fragmentation theorem is understood in its standard nonseparable form. Then check the two short definition bridges: translating a small-diameter relatively weak-open piece into B_X , and using a two-point set in the common kernel to refute every possible strong witness neighbourhood. If those checks pass, the argument yields the exact finite-dimensional characterization claimed in Theorem 2.

References

- [1] D. Ravasini, *Compactivorous Sets in Banach Spaces*, Proc. Amer. Math. Soc. **150** (2022), 2121–2129; arXiv:2104.02695, doi:10.1090/proc/15851.
- [2] J. E. Jayne, I. Namioka, and C. A. Rogers, *Topological Properties of Banach Spaces*, Proc. London Math. Soc. (3) **66** (1993), 651–672, doi:10.1112/plms/s3-66.3.651.